1 kit

Stomp Rocket Kit

WHAT'S IN THE KIT

- 4 stomp rocket launchers
- 12 pre-made stomp rockets

QUICK-START GUIDE

1. Outside or gym only — these really fly.
2. Angle the launcher, load a rocket, stomp!
3. Never look down the launch tube; set a launch-line rule.
4. Measure distances with chalk or cones for instant data.

FULL LESSONS IN THIS GUIDE

PRO TIP: *Paper rocket templates + the launchers = a design-test-iterate unit that costs almost nothing.*

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LESSON · GRADES K-2

Stomp & Measure

LEARNING OUTCOMES

- Students will measure distances with standard and non-standard units
- Students will connect force input to motion output
- Students will record and compare simple data

MATERIALS

- Stomp rocket launchers + rockets
- Open field or gym
- Cones and meter sticks; sidewalk chalk
- Simple recording sheet (big stomp / little stomp)

INSTRUCTIONS

1. Safety lines first: launchers point downrange, everyone behind the line.
2. Demo one launch; let the gasps settle.
3. Each student launches twice: one gentle stomp, one big stomp.
4. Measure both flights — footsteps first, then meter sticks.
5. Record pairs on the sheet; compare as a class.
6. Circle: what made the difference? Where did the 'push' come from?

ACTIVITY

> Stomp & Measure — big stomps versus little stomps, measured in footsteps and meters. Cause-and-effect, measurement practice, and cardio in one loud, joyful field session.

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LESSON · GRADES 3-5

Launch Angle Lab

LEARNING OUTCOMES

- Students will run a controlled experiment with one variable
- Students will collect multi-trial data and compute averages
- Students will graph results and identify an optimum

MATERIALS

- Stomp rocket launchers + rockets
- Angle guides (30°, 45°, 60° — protractor + cardboard jig)
- Measuring tapes and cones
- Trial data table (3 trials per angle)

INSTRUCTIONS

1. Frame the question: which launch angle flies farthest?
2. Fair-test rules: same stomper per team, same rocket, same effort cue.
3. Three trials at each angle; measure and record every flight.
4. Compute averages per angle — outliers spark good conversations.
5. Graph angle vs. average distance; find the peak.
6. Reveal that ~45° is the physics answer — did their data agree? Why or why not?

ACTIVITY

> Launch Angle Lab — a genuine controlled experiment where the 45° optimum emerges from their own averaged data. The outlier flights teach as much as the clean ones.

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LESSON · GRADES 6-8

Design Your Own Rocket

LEARNING OUTCOMES

- Students will design a rocket varying fins, nose cone, and body
- Students will isolate design variables across flight trials
- Students will connect design features to flight stability and distance

MATERIALS

- Stomp launchers
- Paper rocket templates + tape, scissors
- Fin and nose cone material options
- Flight log: design specs + distances per trial

INSTRUCTIONS

1. Teach the base build: paper tube sized to the launcher, sealed nose.
2. Round 1: everyone flies the identical base rocket for a baseline.
3. Design round: teams modify ONE variable — fin count, fin shape, nose cone, or length.
4. Three trials per design; log specs and distances.
5. Second modification allowed only after data review.
6. Final fly-off plus a claims session: which feature mattered most, and what's your evidence?

ACTIVITY

Design Your Own Rocket — baseline first, one variable at a time, evidence required for claims. A full design-test-iterate unit that costs paper, tape, and enthusiasm.